```
!pip install opencv-python
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
```

```
import numpy as np
import cv2
import matplotlib.pyplot as plt
from google.colab import files
```

```
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving left.png to left (1).png
```

```
image_path = list(uploaded.keys())[0]
```

```
image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
```

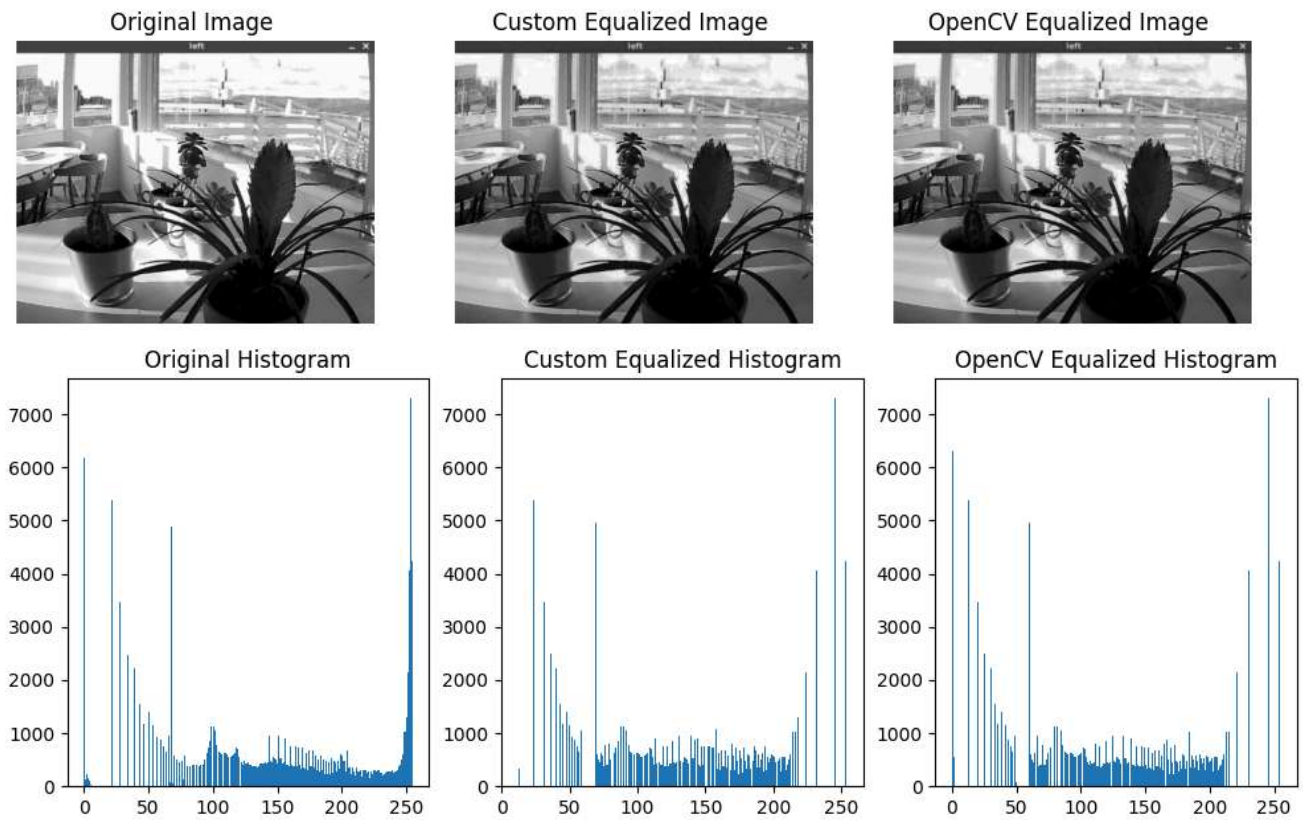
```
if image is None:
    print("Error: Image not loaded.")
    exit()
```

```
def equalize_hist(image, L=256):
    intensity_levels, counts = np.unique(image, return_counts=True)
    total_pixels = image.size
    pdf = counts / total_pixels
    cdf = np.cumsum(pdf)
    transform = np.floor((L - 1) * cdf).astype(np.uint8)
    mapping = dict(zip(intensity_levels, transform))
    equalized_image = np.vectorize(mapping.get)(image)
    return equalized_image.astype(np.uint8)
```

```
custom_image = equalize_hist(image)
opencv_image = cv2.equalizeHist(image)
```

```
plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(custom_image, cmap='gray')
plt.title("Custom Equalized Image")
plt.axis("off")

plt.subplot(1, 3, 3)
plt.imshow(opencv_image, cmap='gray')
plt.title("OpenCV Equalized Image")
plt.axis("off")

plt.show()

plt.figure(figsize=(12, 4))

plt.subplot(1, 3, 1)
plt.hist(image.ravel(), bins=256)
plt.title("Original Histogram")

plt.subplot(1, 3, 2)
plt.hist(custom_image.ravel(), bins=256)
plt.title("Custom Equalized Histogram")

plt.subplot(1, 3, 3)
plt.hist(opencv_image.ravel(), bins=256)
plt.title("OpenCV Equalized Histogram")
```

```
plt.show()
```



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